



The business problem comes first

How can the store reduce peak-hour queue time while protecting service, cost, and risk?

THE PROBLEM

The queue is too long.

- Average peak-hour queue: 7.5 minutes.
- The store handles 84 transactions per hour.
- Customers still need accessible service.

DO NOT JUMP TO ONE ANSWER

Several methods may help.

- Keep a staffed lane open.
- Add clearer kiosk instructions.
- Place an employee near the kiosks.
- Test two kiosks in a small pilot.

HOW COT HELPS

Compare methods before choosing one.

- Separate facts from assumptions.
- Use the same criteria for each method.
- Test the most promising option before scaling.

What CoT does in this class

CoT is a structured decision rationale. It helps students inspect the basis for a recommendation.

START WITH A QUESTION

State the business problem.

- What must improve?
- What constraints matter?
- What counts as success?

ASK FOR A STRUCTURE

Make the decision logic visible.

- Facts provided.
- Assumptions to test.
- Options and trade-offs.

USE THE RESULT CAREFULLY

CoT is not proof.

- Check facts against the dataset.
- Ask for missing information.
- Test before scaling.

Use a four-step decision frame

The model should organize a decision. It should not silently invent the brief.

1. Define

Decision and success metric.

2. Diagnose

Facts, unknowns, assumptions.

3. Compare

Options on the same criteria.

4. Test

Pilot, target, and stop rule.

Bad prompt: no decision frame

The question sounds clear, but it does not define the business problem.

PROMPT TO AVOID

Should our grocery store
replace cashiers with
self-checkout kiosks?

WHAT IS MISSING?

- Store facts and budget.
- A success metric.
- A limit on assumptions.

Bad prompt: false certainty

Do not ask the model to guarantee a forecast that your data cannot support.

PROMPT TO AVOID

We have 12 employees.
Tell us exactly how much
profit two kiosks will add.
Guarantee customer acceptance.

WHY IT FAILS

- No cost or traffic data.
- Guarantees are impossible.
- It rewards invented numbers.

Good prompt: diagnose before deciding

Copy this prompt when you need to turn an unclear question into a research plan.

COPY AND ADAPT

Supermarket case

Known: 12 employees

Goal: shorter peak queues

Risks: theft, downtime, access

Make three lists:

1. Facts
2. Assumptions to test
3. Missing data

Do not invent cost estimates.

How to audit the response

Before accepting the output, ask whether every claim has a place in your evidence.

CHECK 1

Facts

- Can you point to a source for each fact?

CHECK 2

Assumptions

- Can each assumption be tested before rollout?

CHECK 3

Unknowns

- Did the answer clearly request missing data?

Session 1.4 self-checkout pilot data

Use this fictional instructional dataset in the next prompt and exercises. The CSV file contains the same values.

DEMAND

Peak period

- 84 transactions per hour.
- 7.5-minute average queue.
- 20% of shoppers aged 65 or older.

COST

Proposed pilot

- Two self-checkout kiosks.
- \$1,150 monthly lease per kiosk.
- \$250 monthly support cost.

RISK AND TARGET

90-day guardrails

- 1.2% baseline shrinkage.
- Target: 5-minute queue and 98% uptime.
- Stop at shrinkage above 1.4%.

Good prompt: compare options

The data are already provided. Do not invent values that are absent from the case.

COPY AND ADAPT: PART 1

Use only the Session 1.4
self-checkout pilot data.

Compare three options:

- current checkout
- pilot two kiosks for 90 days
- expand all lanes

Do not invent missing values.

Good prompt: make the pilot decision

This second section turns the comparison into an action that can be stopped or scaled.

COPY AND ADAPT: PART 2

Use these criteria:

- customer wait time
- total operating cost
- shrinkage risk
- staff workload
- service access

Recommend one option.

State one pilot target
and one stop rule.

Say "Information needed"
when evidence is missing.

Exercise 1: Assumption audit

Use the data on Slide 7 and the CSV file. Submit one audit table, not a generic recommendation.

STEP 1

Write the brief

- Decision owner.
- Deadline.
- Success metric.

STEP 2

Run Good Example 1

- List facts.
- Mark assumptions.
- List unknowns.

DELIVERABLE

One-page audit

- Five questions for the owner.
- One source for each fact.

Exercise 2: Simple pilot review

After 90 days, the average peak-hour queue is 6.5 minutes. The target was five minutes or less.

STEP 1

Name the problem

- Queue time is above target.
- Customers need help at busy times.

STEP 2

Choose one action

- Place one employee near kiosks.
- Add a short instruction sign.
- Keep one staffed lane open.

STEP 3

Measure again

- Track queue time for two weeks.
- Target: five minutes or less.
- State whether the action worked.

Before you accept an AI recommendation

A recommendation is useful only when another person can check its evidence and conditions.

FRAME

Define success

- Owner, deadline, and metric.

PROTECT

Label uncertainty

- Facts, assumptions, and unknowns.

TEST

Pilot before scale

- Target, stop rule, and review owner.